

The Fundamental Theorem of Asset Pricing, Formalized in Lean 4

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Abstract

The Fundamental Theorem of Asset Pricing states that a market is free of arbitrage exactly when it admits an equivalent martingale measure. We formalize it in Lean 4 over Mathlib in three settings: a finite-state market with one risky asset over a finite horizon (Harrison–Pliska), a one-period market with one risky asset on an arbitrary probability space, and a one-period market with finitely many assets on an arbitrary probability space. The finite case is the geometry of a separating hyperplane; the scalar one-period case is an elementary change of density. In the d -asset case the proof builds the equivalent martingale measure from a minimiser of the potential $\theta \mapsto \mathbb{E}[\log(1 + e^{\langle \theta, Y \rangle})]$: absence of arbitrage makes the potential coercive off the redundant directions, its first-order condition is the martingale property, and for integrable returns the logistic weight of the minimiser, normalised, is the density; a bounded tempering removes the integrability assumption, so the return need only be measurable. The construction uses no Hahn–Banach theorem, no L^0 -closedness argument, no measurable selection and no non-redundancy hypothesis. The separation lemma behind the finite case has a companion, built from the same two helper lemmas, that yields the Artzner–Delbaen–Eber–Heath representation of coherent risk measures on a finite state space. The boundary is explicit. The general multi-period theorem of Dalang, Morton and Willinger is formalized only in its forward direction, and so is the continuous-time theorem, for simple strategies. Every theorem is sorry-free, the axioms of each result discussed are pinned to Lean’s three standard axioms by a build-enforced gate, and the development is reproducible from a pinned toolchain.

1 INTRODUCTION

A market offers an *arbitrage* if some admissible strategy turns no initial wealth into a payoff that is non-negative almost surely and strictly positive with positive probability. The Fundamental Theorem of Asset Pricing identifies the absence of such opportunities with a single analytic object: a probability measure Q , equivalent to the physical measure P , under which discounted asset prices are martingales. No arbitrage holds if and only if such an *equivalent martingale measure* exists. The equivalence was made precise by Harrison and Kreps [13] and Harrison and Pliska [14], established for general discrete-time models by Dalang, Morton and Willinger [7], and carried to continuous time in the no-free-lunch form of Delbaen and Schachermayer [9]. It is the result on which arbitrage-free pricing rests: a price is the Q -expectation of a discounted payoff.

The theorem is also a natural test for formal mathematics, because its content is analytic rather than computational. In the finite case it hides a separating-hyperplane argument;

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in general it asserts the existence of a change of measure that is usually obtained non-constructively, by separating a cone in L^1 with a Hahn–Banach functional after proving the cone closed in L^0 . Fixing exactly which hypotheses each direction needs, and checking the measure-theoretic details against a kernel, is a task at which formalization is decisive.

Formalized probability has grown substantially, and with it the measure theory on which pricing depends. Echenim, Guiol and Peltier verified Cox–Ross–Rubinstein pricing in Isabelle/HOL [11], Keskin formalized discrete- and continuous-time martingales [17], Avigad, Hölzl and Serafin the central limit theorem [2], and Degenne and collaborators constructed Brownian motion in Lean [8, 3]. The nearest precedent for the present paper is Nagy’s Lean 4 derivation from Itô’s lemma to Black–Scholes [19], which contains the theorem for a one-period market with one asset and two states. A search in October 2026 of the Lean ecosystem, the Isabelle Archive of Formal Proofs and the Coq/Rocq ecosystem found no machine-checked proof beyond that case: none on an arbitrary probability space, none with several assets on an arbitrary space, and none over several periods.

What this paper does. We formalize the theorem in three settings of increasing generality and, in the most general of them, give the equivalent martingale measure by a formula.

- (1) *Finite states, finitely many periods* (Section 3). For one risky asset, the backward direction is the geometry of a hyperplane separating the attainable gains from the standard simplex (ftap_discrete); in one period the same separation handles finitely many assets.
- (2) *One period, one asset, arbitrary probability space* (Section 4). Integrability is no longer free, and the measure is built by an elementary change of density (ftap_one_period).
- (3) *One period, d assets, arbitrary probability space* (Section 5). For integrable returns the proof takes as density of the equivalent martingale measure the normalised logistic weight of a minimiser of the softplus potential $\mathbb{E}[\log(1 + e^{\langle \theta, Y \rangle})]$; a bounded tempering removes integrability. Redundant assets are absorbed rather than excluded, and the return need only be measurable (ftap_one_period_vector).
- (4) *One separation for pricing and risk* (Section 7). The separation lemma of the finite case and the one behind the representation of coherent risk measures on a finite state space (coherentRisk_isLUB) are two faces of one finite-dimensional separation, proved side by side from shared helper lemmas.
- (5) *A precise boundary* (Section 8). The general multi-period theorem and the continuous-time theorem are formalized only in their forward directions; we say exactly what is proved.

This paper is part of a Lean 4 [10] library of formalized mathematical finance built on Mathlib [18]; the library is described in a companion overview [5] and its Itô calculus in a second companion [4]. The results developed here live entirely within Mathlib’s measure theory and convex analysis, on its withDensity change of measure and its finite-dimensional separation theorems; they stand alongside, but do not depend on, the library’s Brownian-motion layer.

Changes from the first version. This version corrects three statements of the first. The two-state case was formalized earlier [19], so the claim of priority is narrowed. The d -asset density is the softplus analogue of the Esscher measure, not the Esscher measure itself (Section 5.4). And only one direction of the correspondence between no arbitrage and coercivity is proved, so the text no longer calls them equivalent. The version adds the

separation shared with coherent risk measures (Section 7), the finite-state d -asset theorem and the forward directions that border the development (Sections 3 and 8), and updates the artifact to the current release (Section 9).

2 THE MARKET AND THE TWO EASY FACTS

In each setting a position in the traded assets produces a discounted gain, and an arbitrage is a position whose gain is non-negative almost surely without being almost surely zero. The settings differ only in what a position is and how its gain is formed.

- In the *finite-state* model a filtration $\mathcal{F} = (\mathcal{F}_t)_{t \leq T}$ on a finite probability space carries an adapted discounted price $S_0, \dots, S_T$ of one risky asset. A strategy φ is predictable, with φ_{t+1} measurable for $\mathcal{F}_t$, and its discounted gain is the martingale transform $(\varphi \cdot S)_T = \sum_{t < T} \varphi_{t+1}(S_{t+1} - S_t)$.
- In the *scalar one-period* model a discounted excess return $Y: \Omega \rightarrow \mathbb{R}$ is held in position $\theta \in \mathbb{R}$, with gain θY .
- In the *d -asset one-period* model the return $Y: \Omega \rightarrow F$ takes values in a finite-dimensional real inner-product space F (the d -asset market is $F = \mathbb{R}^d$), a constant position $\theta \in F$ is held, and the gain is $\langle \theta, Y \rangle$. (In Mathlib the d -asset space is `EuclideanSpace ℝ (Fin d)`, not the product $\mathbb{R}^d$ with its sup norm.)

Write G for the discounted gain of a position. *No arbitrage* says that a gain which is non-negative is null:

$$G \geq 0 \quad P\text{-a.s.} \quad \implies \quad G = 0 \quad P\text{-a.s.},$$

which is the textbook “no $G \geq 0$ with $P(G > 0) > 0$ ” in contrapositive form. The condition is stated under P alone and names no second measure. An *equivalent martingale measure* is a probability measure Q with $Q \ll P$ and $P \ll Q$ under which discounted prices are fair: in the one-period settings Y is Q -integrable with $\mathbb{E}_Q[Y] = 0$ (vector-valued in the d -asset case); in the finite-state setting $S_t = \mathbb{E}_Q[S_{t+1} \mid \mathcal{F}_t]$ for every $t < T$. Recording equivalence as the pair of absolute continuities is what lets a Q -almost-sure statement transfer back to P . Listing 1 shows the d -asset definitions; the other two settings differ only in the gain.

Listing 1. No arbitrage and the equivalent martingale measure, d -asset one-period market (Foundations/FTAPOnePeriodVector).

```
variable {Ω : Type*} {mΩ : MeasurableSpace Ω}
  (P : Measure Ω) [IsProbabilityMeasure P]
  {F : Type*} [NormedAddCommGroup F] [InnerProductSpace ℝ F]
  [FiniteDimensional ℝ F] [MeasurableSpace F] [BorelSpace F] (Y : Ω → F)

def NoArbitrage : Prop :=
  ∀ θ : F, 0 ≤m[P] (fun ω ↦ inner ℝ θ (Y ω)) →
    (fun ω ↦ inner ℝ θ (Y ω)) =m[P] 0

structure IsEMM (Q : Measure Ω) : Prop where
  prob : IsProbabilityMeasure Q
  absP : Q <= P
  Pabs : P <= Q
  int : Integrable Y Q
  fair : ∫ ω, Y ω ∂Q = 0
```

One direction of the theorem is uniform and elementary. If an equivalent martingale measure Q exists, the expected discounted gain of every position vanishes under Q : in the one-period settings $\mathbb{E}_Q[\theta Y] = \theta \mathbb{E}_Q[Y] = 0$ and $\mathbb{E}_Q[\langle \theta, Y \rangle] = \langle \theta, \mathbb{E}_Q[Y] \rangle = 0$, and in the finite-state setting $\mathbb{E}_Q[(\varphi \cdot S)_T] = 0$ because each term $\varphi_{t+1}(S_{t+1} - S_t)$ has zero $\mathcal{F}_t$ -conditional mean. A gain with $G \geq 0$ and $\mathbb{E}_Q[G] = 0$ is zero Q -almost surely, hence P -almost surely by equivalence, so the position is not an arbitrage. This is `noArbitrage_of_isEMM` in each development. The content of the theorem is the converse, which the next three sections supply in rising generality.

3 FINITE MARKETS: THE GEOMETRIC THEOREM

In a finite-state market the equivalent martingale measure is, almost literally, a separating hyperplane. Fix a finite probability space (Ω, P) with full support, a filtration $\mathcal{F}$, an adapted scalar discounted price S and a horizon T . The discounted gains attainable by predictable strategies form a linear subspace of $\mathbb{R}^\Omega$, the span of the one-period gains $\mathbf{1}_A(S_{t+1} - S_t)$ with $t < T$ and $A \in \mathcal{F}_t$; call it the *gains subspace*. No arbitrage says exactly that this subspace meets the non-negative cone only at the origin, equivalently that it is disjoint from the standard simplex of probability vectors on Ω .

The simplex is compact and convex and the gains subspace is closed, being finite-dimensional, so the geometric Hahn–Banach theorem separates them strictly. The library takes this one level higher, in a form that serves more than pricing. Its root ([Listing 2](#)) is stated for a closed convex *cone* $C \subseteq \mathbb{R}^l$ disjoint from the simplex, and produces a strictly positive functional q that is non-positive on C . A subspace is a cone closed under negation, so applied to the gains subspace the one-sided bound becomes two-sided: q annihilates every attainable gain (`exists_pos_dual_of_disjoint_stdSimplex`). Read as a measure, q makes every martingale transform have zero expectation.

Listing 2. The cone-separation root (Foundations/ConvexDuality).

```
theorem exists_pos_separating_of_cone_disjoint_simplex
  {ι : Type*} [Fintype ι]
  (C : Set (ι → ℝ)) (hCconv : Convex ℝ C) (hCclosed : IsClosed C)
  (hCcone : ∀ {x}, x ∈ C → ∀ {c : ℝ}, 0 ≤ c → c • x ∈ C)
  (hdisj : Disjoint (stdSimplex ℝ ι) C) :
  ∃ q : ι → ℝ, (∀ i, 0 < q i) ∧ ∀ v ∈ C, ∑ i, q i * v i ≤ 0
```

The cone form is not needed for the market, whose gains form a subspace; it is stated for the sake of [Section 7](#), where a genuine cone appears. The market reaches the theorem in two steps. Under no arbitrage the gains subspace misses the simplex (`gains_disjoint_stdSimplex`): a non-negative gain is null, and on a full-support space a null vector has no positive coordinate, while every point of the simplex has one. The strictly positive annihilator, normalised to a probability vector, is a measure $Q \sim P$; it annihilates each generator $\mathbf{1}_A(S_{t+1} - S_t)$, which is the defining property of the conditional expectation, so S is a Q -martingale up to T . With the forward direction of [Section 2](#) this is the biconditional ([Listing 3](#)).

Listing 3. The finite-state theorem (Foundations/FTAPDiscrete). Ω is a Fintype with `MeasurableSingletonClass`, $\mathcal{F}$ a filtration indexed by $\mathbb{N}$, $S : \mathbb{N} \rightarrow \Omega \rightarrow \mathbb{R}$.

```
theorem ftap_discrete (hS : StronglyAdapted ℱ S) (hP : ∀ ω, 0 < P {ω}) :
  NoArbitrage ℱ P S T ↔ ∃ Q, IsEMM ℱ P S T Q
```

This is the theorem of Harrison and Pliska, the finite case of Dalang, Morton and

Willinger. The statement is slightly stronger than the textbook one: P need only be a measure of full support on the finite space, not a probability measure, and strategies are unbounded. The same kernel gives the one-period theorem for finitely many assets on a finite state space: the gains subspace is then the span of the assets' excess-return vectors, and the statement is the biconditional `hasEMM_multi_iff_not_hasArbitrage` (`Foundations/FTAPMultiState`). The multi-period theorem is stated for one risky asset; with several assets only the generators of the gains subspace would change, but that version is not in the library. The restriction to finitely many states is what keeps the separating functional elementary and the measure a genuine probability vector; the next two sections give it up.

4 ONE PERIOD, ONE ASSET: THE ELEMENTARY THEOREM

Leaving finitely many states changes the problem. On an arbitrary probability space the attainable gains need not form a closed set, and the separation argument of [Section 3](#) has nothing compact to grip. For a single scalar return, though, the equivalent martingale measure can still be written down directly, with no separation at all.

Fix a measurable $Y: \Omega \rightarrow \mathbb{R}$ on an arbitrary (Ω, P) . Two difficulties replace the single one of the finite case. First, Y need not be integrable, so $\mathbb{E}_Q[Y]$ may fail even to make sense. Second, a fair measure must be produced from P rather than chosen from a simplex. Both yield to changes of density.

Integrability is bought by *tempering*. The bounded strictly positive weight $w = (1 + |Y|)^{-1}$, normalised by $\kappa = \mathbb{E}[w] \in (0, 1]$, defines $\tilde{P} = w/\kappa \cdot P$, equivalent to P , under which Y is integrable because $|Y|w/\kappa \leq \kappa^{-1}$. No arbitrage is an almost-sure notion, so it survives the passage to the equivalent $\tilde{P}$. Over $\tilde{P}$ the construction is a dichotomy on the sign of Y : if $Y \geq 0$ almost surely and is not almost surely zero then $\theta = 1$ is an arbitrage, and symmetrically for $Y \leq 0$, so under no arbitrage the positive and negative parts of Y both carry mass. A two-region density that re-weights the two sides,

$$Z = \lambda \mathbf{1}_{\{Y \geq 0\}} + \mu \mathbf{1}_{\{Y < 0\}}, \quad \lambda, \mu > 0,$$

can then be balanced so that $\mathbb{E}[ZY] = 0$ and normalised so that $\mathbb{E}[Z] = 1$; one choice of λ, μ , explicit in the two tail integrals of Y , does both at once. If $Y = 0$ almost surely, $Q = P$ will do. The balancing core is `exists_isEMM_of_pos_tails`; removing the integrability assumption by the tempering above gives `exists_isEMM_of_noArbitrage`, whose measure has P -density proportional to $Z(1 + |Y|)^{-1}$, with Z computed under $\tilde{P}$. The forward direction closes the biconditional ([Listing 4](#)).

Listing 4. The scalar one-period theorem (`Foundations/FTAPOnePeriod`); $Y: \Omega \rightarrow \mathbb{R}$ on an arbitrary probability space.

```
theorem ftap_one_period (hY : Measurable Y) :
  NoArbitrage P Y  $\leftrightarrow$   $\exists$  Q, IsEMM P Y Q
```

This is the one-asset case of the one-period theorem of Dalang, Morton and Willinger, in the form found in Föllmer and Schied [12, Ch. 1]. No Hahn–Banach theorem and no Kreps–Yan argument enter; the measure is built rather than selected. What the argument leaves open is dimension. A single asset has only two sides to balance, and the dichotomy that drives the construction does not by itself reach a market of several assets.

5 ONE PERIOD, MANY ASSETS: THE VARIATIONAL CONSTRUCTION

The d -asset one-period theorem is where the construction leaves the textbook proof furthest behind. The return is now $Y: \Omega \rightarrow F$ with F a finite-dimensional inner-product

space, and a position is a constant vector $\theta \in F$. The classical route separates the cone of attainable claims from the non-negative payoffs in L^1 by a Hahn–Banach functional, an argument whose infinite-dimensional duality and cone-closedness are heavy to formalize. We replace it: the equivalent martingale measure is the minimiser of one smooth convex function on the finite-dimensional space F , and every step is a finite-dimensional or elementary-integral fact. Throughout, the Lean spelling of the gain $\langle \theta, Y \rangle$ is `inner ℝ θ (Y ↦ ω)`.

5.1 THE POTENTIAL

For $\theta \in F$ set

$$f(\theta) = \mathbb{E}[\log(1 + e^{\langle \theta, Y \rangle})],$$

the expected *softplus* of the gain. The softplus $u \mapsto \log(1 + e^u)$ is smooth and convex, with derivative the logistic function $\sigma(u) = e^u / (1 + e^u) \in (0, 1)$, and it satisfies

$$u^+ \leq \log(1 + e^u) \leq |u| + \log 2.$$

The upper bound dominates the integrand by $\|\theta\| \|Y\| + \log 2$, so f is finite and continuous whenever Y is integrable; the lower bound will give coercivity. Because σ is bounded, differentiation under the integral sign is dominated by $\|u\| \|Y\|$, and the derivative in a direction u is

$$\left. \frac{d}{dt} \right|_{t=0} f(\theta + tu) = \mathbb{E}[\sigma(\langle \theta, Y \rangle) \langle u, Y \rangle] \quad (\text{hasDerivAt_potential_dir}),$$

the directional form of the gradient $\mathbb{E}[\sigma(\langle \theta, Y \rangle) Y]$, the Y -average weighted by the logistic of the current gain (Listing 5). The potential is also convex, but the proof never uses that: existence of a minimiser needs only continuity and coercivity, and the first-order condition holds at any minimiser.

Listing 5. The softplus potential (Foundations/FTAPOnePeriodVector).

```
noncomputable def softplus (u : ℝ) : ℝ := Real.log (1 + Real.exp u)
noncomputable def logistic (u : ℝ) : ℝ := Real.exp u / (1 + Real.exp u)
noncomputable def potential (θ : F) : ℝ :=
  ∫ ω, softplus (inner ℝ θ (Y ω)) ∂P
```

5.2 NO ARBITRAGE GIVES COERCIVITY

No arbitrage is what makes f attain a minimum. Redundant directions are set aside first. The *gains kernel*

$$N = \{\theta \in F : \langle \theta, Y \rangle = 0 \text{ } P\text{-a.s.}\}$$

collects the positions with almost surely zero gain. It is a linear subspace, and f is constant along it, since adding an element of N changes no gain. On the orthogonal complement $N^\perp$ the picture is rigid. The positive-gain average $g(\theta) = \mathbb{E}[\langle \theta, Y \rangle^+]$ is continuous and positively homogeneous, and under no arbitrage it is strictly positive on $N^\perp \setminus \{0\}$: if $g(\theta) = 0$ then $\langle -\theta, Y \rangle \geq 0$ almost surely, so no arbitrage puts θ in N , while $N \cap N^\perp = \{0\}$. Its minimum c over the unit sphere of $N^\perp$ is therefore positive, and since the softplus dominates the positive part,

$$f(\theta) \geq g(\theta) \geq c \|\theta\| \quad (\theta \in N^\perp).$$

No arbitrage makes f coercive on $N^\perp$ (Listing 6). The converse holds as well, though the theorem does not need it and it is not formalized: if θ is an arbitrage, then at every point η the derivative of f in the direction $-\theta$ is $-\mathbb{E}[\sigma(\langle \eta, Y \rangle) \langle \theta, Y \rangle] < 0$, so f has no minimiser.

Listing 6. Coercivity from no arbitrage (Foundations/FTAPOnePeriodVector).

```
lemma exists_pos_lower_bound (hYint : Integrable Y P) (hNA : NoArbitrage P Y)
  (hNbot : (gainsKernel P Y)^\perp \neq \perp) :
  \exists c > 0, \forall \theta \in (gainsKernel P Y)^\perp, c * \|\theta\| \leq potential P Y \theta
```

5.3 THE MINIMISER AND ITS MEASURE

Coercivity and continuity give a minimiser θ_0 of f on the closed subspace $N^\perp$, by compactness of a closed ball (exists_global_min_potential); because f is flat along N and $F = N \oplus N^\perp$, the same θ_0 minimises f over all of F . At a global minimiser every directional derivative vanishes, so

$$\mathbb{E}[\sigma(\langle \theta_0, Y \rangle) Y] = 0$$

(Listing 7). The weight $z = \sigma(\langle \theta_0, Y \rangle)$ lies strictly between 0 and 1: it is bounded, strictly positive, and makes Y fair. The measure Q with P -density $z/\mathbb{E}[z]$ is therefore a probability measure equivalent to P , under which Y is integrable with $\mathbb{E}_Q[Y] = 0$. If $N^\perp$ is trivial, every direction is redundant, $Y = 0$ almost surely, and $Q = P$ will do.

Listing 7. The first-order condition (Foundations/FTAPOnePeriodVector).

```
lemma integral_logistic_smul_eq_zero (hY : Measurable Y) (hYint : Integrable Y P)
  {\theta_0 : F} (hmin : \forall \theta, potential P Y \theta_0 \leq potential P Y \theta) :
  \int \omega, logistic (inner \mathbb{R} \theta_0 (Y \omega)) \bullet Y \omega \partial P = 0
```

Two features set this apart from the textbook proof. No non-redundancy hypothesis is needed: redundant assets live in N , the minimisation takes place on $N^\perp$, and the kernel is absorbed rather than assumed away. And integrability of Y is not assumed: as in Section 4, the bounded tempering $(1 + \|Y\|)^{-1}$ reduces to the integrable case, and equivalence of measures carries the result back. The minimisation then takes place under the tempered measure $\tilde{P}$, so in general the density of Q with respect to P is proportional to $\sigma(\langle \theta_0, Y \rangle) (1 + \|Y\|)^{-1}$, with θ_0 a minimiser of the potential under $\tilde{P}$. The backward direction is exists_isEMM_of_noArbitrage; with the forward direction it is the biconditional (Listing 8).

Listing 8. The d -asset one-period theorem (Foundations/FTAPOnePeriodVector); the context is that of Listing 1.

```
theorem ftap_one_period_vector (hY : Measurable Y) :
  NoArbitrage P Y \leftrightarrow \exists Q, IsEMM P Y Q
```

Where the classical proof invokes the dual of L^∞ and the closedness of a cone in L^0 , this one invokes coercivity, a minimum over a sphere, and differentiation under an integral, each already in Mathlib. The reward is an equivalent martingale measure given by a formula in θ_0 , $\sigma(\langle \theta_0, Y \rangle)$ up to normalisation and tempering, rather than produced by a separating functional. The minimiser itself is obtained by compactness, not computed; “explicit” here means a formula in θ_0 , not a closed form. The formula lives in the proof: the theorem asserts existence, and what the library states about the formula is the first-order condition of Listing 7, from which the martingale property of Q follows by normalisation.

5.4 WHY THE SOFTPLUS

Variational proofs of the one-period theorem are not new: Rogers [20] obtains the equivalent martingale measure from a minimisation argument rather than a separation. What matters for a formalization is the choice of potential, and the softplus is chosen for what the argument asks of it. For a convex C^1 function ϕ in place of the softplus, the proof uses four properties:

- (i) $\phi(u) \geq u^+$, which turns no arbitrage into coercivity;
- (ii) $|\phi(u)| \leq a|u| + b$, which makes $\mathbb{E}[\phi(\langle \theta, Y \rangle)]$ finite for integrable Y ;
- (iii) ϕ' bounded, which dominates the derivative under the integral by an L^1 function;
- (iv) $\phi' > 0$, which makes the weight $\phi'(\langle \theta_0, Y \rangle)$ a strictly positive density.

The softplus has all four, with $\phi' = \sigma \in (0, 1)$. The exponential $\phi(u) = e^u$ has (i) and (iv) but neither (ii) nor (iii): its minimiser gives the Esscher density proportional to $e^{\langle \theta_0, Y \rangle}$, which needs exponential moments of Y , and the bounded tempering $(1 + \|Y\|)^{-1}$ supplies only integrability. The logistic density is thus the softplus analogue of the Esscher measure, not the Esscher measure itself (the first version of this paper conflated the two). It is bounded, and it exists whenever Y is integrable.

REMARK 5.1 (interpretation, not formalized). The convex conjugate of the softplus is $q \log q + (1 - q) \log(1 - q)$ on $[0, 1]$, minus the binary entropy $H(q)$. Interchanging the minimum over θ with the supremum over weights, which convex duality justifies but we do not formalize, identifies $\min_{\theta} f(\theta)$ with $\max\{\mathbb{E}[H(z)] : 0 \leq z \leq 1, \mathbb{E}[zY] = 0\}$, attained at $z = \sigma(\langle \theta_0, Y \rangle)$. The density the construction produces is, up to normalisation, the fair weight of maximal expected binary entropy.

6 A WORKED EXAMPLE

A two-state market makes the construction concrete and shows the gains kernel at work. The example is computed by hand; it illustrates the Lean argument and is not itself a Lean theorem. Let $\Omega = \{+, -\}$ with $P(+) = P(-) = \frac{1}{2}$, and take two assets whose discounted excess returns are

$$Y_1(+) = a, \quad Y_1(-) = -b \quad (a, b > 0), \quad Y_2 = c Y_1 \quad (c \neq 0),$$

so the second asset is redundant. A position $\theta = (\theta_1, \theta_2)$ has gain $\langle \theta, Y \rangle = (\theta_1 + c \theta_2) Y_1$, which depends on θ only through the scalar $s = \theta_1 + c \theta_2$. The gains kernel is the line $N = \{\theta : \theta_1 + c \theta_2 = 0\}$, and the potential collapses to a function of s alone,

$$f(\theta) = \frac{1}{2} \log(1 + e^{sa}) + \frac{1}{2} \log(1 + e^{-sb}).$$

Its derivative in s is $\frac{1}{2}(a \sigma(sa) - b \sigma(-sb))$, strictly increasing in s , negative for $s \rightarrow -\infty$ and positive for $s \rightarrow +\infty$, so it vanishes at a unique s_0 . The logistic weight $z = \sigma(s_0 Y_1)$, normalised, is the density, and the first-order condition $a \sigma(s_0 a) = b \sigma(-s_0 b)$ is $\mathbb{E}_Q[Y_1] = 0$. In two states that pins the measure to

$$Q(+) = \frac{b}{a+b}, \quad Q(-) = \frac{a}{a+b},$$

the classical risk-neutral probability, which is also the explicit weight of the two-state module (`emmWeightUp`, after [19]). The redundant asset adds no constraint: once Y_1 is fair,

$Y_2 = c Y_1$ is fair automatically, as the minimisation over $N^\perp$ predicts. In higher dimension the first-order condition is no longer a scalar equation, but the structure is the same: the kernel absorbs the redundant directions, and for integrable returns the logistic of the minimising gain is the density.

7 ONE SEPARATION FOR PRICING AND RISK

The root of [Listing 2](#) has a sibling. A single point x_0 outside a closed convex cone C that contains the origin is separated from it by a functional p with $\sum_i p_i v_i \leq 0$ on C and $\sum_i p_i x_{0,i} > 0$ (`exists_separating_of_not_mem_cone`). Both are derived in the same file from Mathlib’s geometric Hahn–Banach theorems through two shared lemmas, a basis expansion of a functional and the fact that a functional bounded below on a cone is non-negative on it. The first is the pricing reading of separation; the second is the risk reading.

Risk. A risk measure ρ on a finite state space is *coherent* if it is monotone, cash-invariant, positively homogeneous and subadditive [1]. Its acceptance set $\{X : \rho(X) \leq 0\}$ is then a closed convex cone, and a position that is not acceptable is a point outside it. Separating the rejected position $X + (\rho(X) - \varepsilon)1$ from the acceptance cone and normalising the separating functional yields a probability vector that prices every acceptable position non-negatively; letting $\varepsilon \downarrow 0$ gives the representation of Artzner, Delbaen, Eber and Heath ([Listing 9](#)): $\rho(X)$ is the least upper bound of the expected losses $\mathbb{E}_q[-X]$ over the representing set.

Listing 9. The coherent-risk representation on a finite state space (RiskMeasures/AcceptanceSet).

```
def representingSet {ι : Type*} [Fintype ι] (ρ : (ι → ℝ) → ℝ) :
  Set (ι → ℝ) :=
  {q | (∀ i, 0 ≤ q i) ∧ (∑ i, q i = 1) ∧
    ∀ Z ∈ {Y : ι → ℝ | ρ Y ≤ 0}, 0 ≤ ∑ i, q i * Z i}

theorem coherentRisk_isLUB {ι : Type*} [Fintype ι] {ρ : (ι → ℝ) → ℝ}
  (hρ : IsCoherentRisk ρ) (X : ι → ℝ) :
  IsLUB ((fun q ↦ ∑ i, q i * (- X i)) '' representingSet ρ) (ρ X)
```

So the equivalent martingale measure of a finite market and the representing scenarios of a coherent risk measure are the same kind of object, produced the same way: a strictly positive (respectively non-negative) normalised functional that separates a cone. In the library this is more than an analogy. Both theorems import the one module, and the two separations they rest on, cone from simplex and point from cone, are proved from the same two helper lemmas; they are siblings, not one lemma applied twice. The statement of `coherentRisk_isLUB` is also restated with Mathlib alone and checked against the library by an external tool ([Section 9](#)).

Superhedging. Superhedging shows where the separation has not yet been carried. Weak duality needs none: every martingale measure, equivalent or not, prices a claim below the cost of any portfolio that dominates it (`emm_le_superReplication`), by a direct calculation. Strong duality, that the superhedging price equals the supremum of martingale prices, is the separation-based converse. It needs the cone of super-replicable payoffs to be closed, which for a finitely generated cone is classical but is not available in Mathlib at the library’s pin; Mathlib’s cone-separation theorem itself is. Strong duality is not formalized.

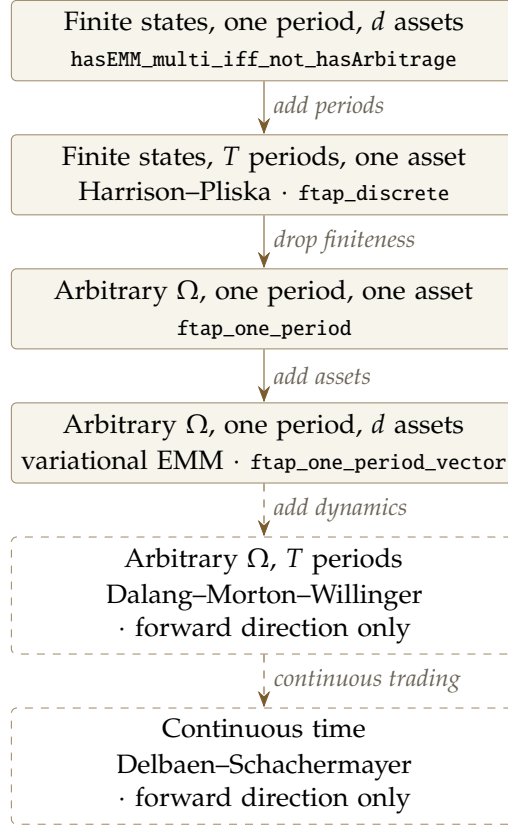


FIGURE 1. The settings formalized here (solid) and those that remain open or partial (dashed). Each solid box is a machine-checked biconditional between no arbitrage and an equivalent martingale measure.

8 WHAT IS PROVED, AND WHAT IS NOT

The theorems above are complete and machine-checked: the finite-state one-period theorem with several assets, the finite-state multi-period theorem with one asset, and the one-period theorem on an arbitrary probability space with one asset and with several. Each is a biconditional between no arbitrage and the existence of an equivalent martingale measure (Figure 1).

The general multi-period theorem. One rung of the ladder is missing its converse. The forward direction holds on an arbitrary probability space over several periods: an equivalent martingale measure excludes arbitrage by bounded predictable strategies in one risky asset (`emm_implies_no_arbitrage`). The theorem of Dalang, Morton and Willinger places a multi-period market on an arbitrary probability space: trading is dynamic, strategies are predictable processes, and the attainable gains live in L^0 rather than in a finite-dimensional space. Its classical proofs need two ingredients that the cases here sidestep: the closedness of the cone of attainable claims in L^0 under no arbitrage, and a measurable selection that assembles the one-period conditional measures into a single density across time [7, 16]. Both are substantial, and neither is formalized; the backward direction is open. So is the finite-state multi-period theorem with several assets. The finite-state case captures the multi-period structure when the sample space is finite; the one-period cases capture the measure-theoretic difficulty in a single step. A natural route to the general theorem through the present construction would apply the softplus minimisation conditionally on $\mathcal{F}_t$, one period at a time; making the conditional minimiser measurable

in ω is where a measurable-selection argument re-enters, and it is not attempted here.

Continuous time. In continuous time the library proves the forward direction for simple strategies: if $Q \sim P$ and the discounted price is a Q -martingale, then no strategy with finitely many trading dates and bounded predictable holdings is an arbitrage (`isEMM_noArbitrageSimple`, `Foundations/ContinuousMarket`). For the Black–Scholes model the equivalent martingale measure is not assumed but constructed, by a Girsanov change of measure [5, 4]. The converse, that no free lunch with vanishing risk yields an equivalent (σ) -martingale measure [9], is not formalized.

The second theorem. Around these results the library carries the uniqueness half of first-course arbitrage theory. In the Brownian market with a driftless price $S = S_0 + \int \sigma dB$, $\sigma \neq 0$ almost everywhere, a probability measure with square-integrable density under which S is a martingale agrees with P on $\mathcal{F}_T^B$ (`measure_eq_of_density`, `Foundations/PricingMeasureL2Density`). Its proof runs through martingale representation and belongs to the Itô calculus [4]. It is the direction from completeness to uniqueness; the converse, which needs the extreme-point characterisation of Jacod and Yor, is not formalized.

9 THE ARTIFACT

The development is part of a Lean 4 library that builds against pinned versions of its dependencies: the Lean toolchain v4.33.0-rc1, Mathlib at revision 0434c03, and the `BrownianMotion` package [3] at 0d5b6eb, which the results of this paper do not use. The state described here is release v1.5.0 (commit c0180c8), archived on Zenodo [21]. A plain lake build from a clean checkout is the canonical check.

Honesty about what the kernel accepts is enforced rather than asserted. Two files run `#print axioms` under `#guard_msgs`: a curated audit of headline results, and a generated one that pins every library constant a benchmark proof cites, resolved by declaration rather than by name. Each theorem discussed here depends only on `propext`, `Classical.choice` and `Quot.sound`; a sorry or an additional axiom anywhere beneath one of them fails the build. A verification ledger records, for every benchmark entry, a hash of the exact sources and toolchain pins it was last checked against, so that an entry whose inputs change cannot keep a stale verdict. For the coherent-risk theorem of Section 7 the repository also carries a Mathlib-only restatement, paired with a witness that its hypothesis is satisfiable, and a proof of both from the library that every build re-checks; the external checker `Comparator` [6] is configured to confirm that the proved statements are exactly the restated ones and that only the three standard axioms are used. The results discussed are collected in Table 1.

Within the wider library these sit among 535 catalogued results: 504 complete formalizations, 18 thin wrappers over Mathlib or `BrownianMotion` theorems, and 13 reduced cores that prove less than the result they are named after, with the 522 complete-or-wrapper entries counting as delivered. The catalogue is published as a dataset in which every entry carries its statement, its status and a description of what its formalization leaves out [15]. The library is released under the Apache 2.0 licence.

Provenance. The Lean development was written by the author with an AI coding assistant (Claude Code), as recorded in the repository’s provenance file; every proof is checked by the Lean kernel, and the gates above apply to it unchanged. This manuscript was revised with the same assistant, and its statements were checked against the Lean sources of release v1.5.0.

Lean declaration	Module	Benchmark entry
<code>MathFin.ftap_discrete</code>	<code>Foundations/FTAPDiscrete</code>	<code>mf-ftap-discrete-complete</code>
<code>MathFin.hasEMM_multi_iff_not_hasArbitrage</code>	<code>Foundations/FTAPMultiState</code>	<code>mf-ftap-single-period-complete</code>
<code>MathFin.OnePeriod.ftap_one_period</code>	<code>Foundations/FTAPOnePeriod</code>	<code>mf-ftap-one-period-general</code>
<code>MathFin.OnePeriodVector.ftap_one_period_vector</code>	<code>Foundations/FTAPOnePeriodVector</code>	<code>mf-ftap-one-period-vector</code>
<code>MathFin.exists_pos_separating_of_cone_disjoint_simplex</code>	<code>Foundations/ConvexDuality</code>	<code>mf-convex-duality-root</code>
<code>MathFin.coherentRisk_isLUB</code>	<code>RiskMeasures/AcceptanceSet</code>	<code>mf-coherent-risk-representation</code>
<code>MathFin.emm_le_superReplication</code>	<code>Foundations/SuperhedgingDuality</code>	<code>mf-superhedging-emm-bound</code>
<code>MathFin.emm_implies_no_arbitrage</code>	<code>Foundations/FTAP</code>	<code>mart-thm-2.6.7</code>
<code>MathFin.ContinuousMarket.isEMM_noArbitrageSimple</code>	<code>Foundations/ContinuousMarket</code>	<code>gir-continuous-emm-forward</code>
<code>MathFin.PricingMeasureL2Density.measure_eq_of_density</code>	<code>Foundations/PricingMeasureL2Density</code>	<code>gir-pricing-measure-density</code>

TABLE 1. The results discussed in this paper, their modules under `MathFin/`, and their benchmark entries. Every declaration is pinned by the axiom audits to `propext`, `Classical.choice` and `Quot.sound`.

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